

Theorems

Part I

Grade 10 Chapter 8 (Triangles) I

Pg. 85 $\rightarrow$ • Theorem: The exterior angle formed when a side of a triangle is produced is equal to the sum of the two interior opposite angles.

Pg 91 $\rightarrow$ • Theorem: The sum of the interior angles of a triangle is 180° .

Pg 99 $\rightarrow$ • Theorem (converse of the theorem on isosceles triangle): The sides opposite equal angles of a $\triangle$ are equal.

Chapter 10 (Triangles II)

Pg 114 $\rightarrow$ • Theorem: If 2 sides are equal in a triangle, the angles opposite the equal sides are equal.

Chapter 16 (parallelograms I)

Pg 164 $\rightarrow$ • Theorem 1: In a parallelogram,

- i) Opposite sides are equal.
- ii) Opposite angles are equal.
- iii) Area of the parallelogram is bisected by each diagonal.

• Theorem 2: In a parallelogram, the diagonals bisect each other.

Chapter 17 (parallelograms II)

Pg 169 $\rightarrow$ • Theorem: If the opposite sides of a quadrilateral are equal, then it is a parallelogram.

Pg 170 $\rightarrow$ • Theorem: If the opposite angles of a quadrilateral are equal, then it is a parallelogram.

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Pg 170 • Theorem: If the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Pg 171 • Theorem: In a quadrilateral, if a pair of opposite sides is equal and parallel, then the quadrilateral is a parallelogram.

Part II

Ch 27 (Chords of a circle)

Pg 102 • Theorem: The straight line joining the centre of the circle to the midpoint of a chord is perpendicular to the chord.

Pg 108 • Theorem: The perpendicular drawn from the centre of a circle to a chord bisects the chord.

Ch 31 (Angles in a circle)

Pg 168 • Theorem: The angle subtended by an arc at the centre of a circle is twice the angle subtended by the same arc on the remaining part of the circle.

Pg 176 • Theorem: The angles in the same segment of a circle are equal.

Pg 182 • Theorem: An angle in a semicircle is a right angle.